Task – 3

Project: Static Timing Analysis on synthesized netlist

1. Objective of the Project

To analyze and ensure timing correctness of a synthesized digital circuit by performing Static Timing Analysis (STA) and fixing any violations.

2. Sample Design Assumptions

(typical for ALU / small datapath)

- Clock frequency = 100 MHz
- Clock period = 10 ns
- Standard cell library = generic .lib

3. SDC File (Timing Constraints)

```
# Create clock (10 ns period → 100 MHz)
create_clock -name clk -period 10 [get_ports clk]
```

```
# Input delay (relative to clock)
set_input_delay 2 -clock clk [all_inputs]

# Output delay
set_output_delay 2 -clock clk [all_outputs]

# Clock uncertainty
set_clock_uncertainty 0.2 [get_clocks clk]

# Set load capacitance (optional)
set_load 0.1 [all_outputs]

# Driving cell (optional)
set_driving_cell -lib_cell INVX1 [all_inputs]
```

4. OpenSTA Script

```
read_liberty typical.lib
read_verilog synthesized_netlist.v
link_design top_module

read_sdc constraints.sdc

# Set operating conditions
set_propagated_clock [all_clocks]

# Run timing checks
report_checks
report_tns
```

```
report_wns
report_timing -max_paths 5
```

5. Sample Timing Report

◈ Setup Analysis

Startpoint: reg_A (rising edge-triggered flip-flop)
Endpoint: reg_B (rising edge-triggered flip-flop)

Path Delay: 11.20 ns
Required Time: 10.00 ns

Slack (setup): -1.20 ns ❌

◈ Hold Analysis

Path Delay: 0.35 ns
Required: 0.20 ns

Slack (hold): +0.15 ns ✅

6. Key Results

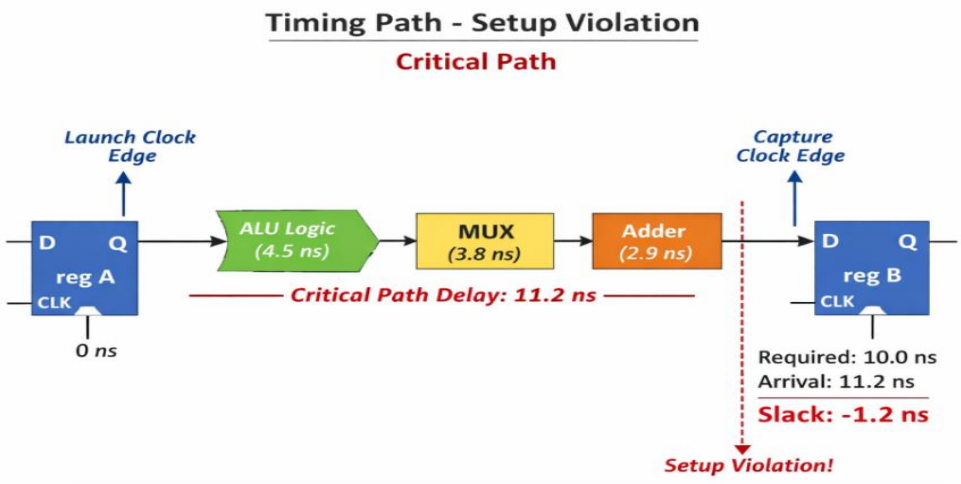
Parameter	Value
Clock Period	10 ns
Worst Negative Slack (WNS)	-1.20 ns ✕
Total Negative Slack (TNS)	-3.50 ns ✕
Hold Violations	None

7. Root Cause of Violation

Setup violation occurs because:-

- Combinational delay too large between flip-flops
- Critical path includes:
  - ALU logic
  - Multiplexer chain
  - Adder carry propagation

8. Timing Path Diagram



9. Fixes

1. Pipelining

Insert registers between stages:-

Before:
Reg → ALU → Reg

After:
Reg → ALU_part1 → Reg → ALU_part2 → Reg

2. Logic Optimization (Re-synthesis)

Run Yosys with better optimization:

```
yosys -p "synth -top top_module; opt; abc -fast"
```

3. Reduce Clock Frequency

Increase clock period:

Old: 10 ns
New: 12 ns

10. Final Result

Static Timing Analysis was successfully performed using OpenSTA. The design shows setup timing violations with WNS = -1.20 ns, indicating that the critical path exceeds the clock period.